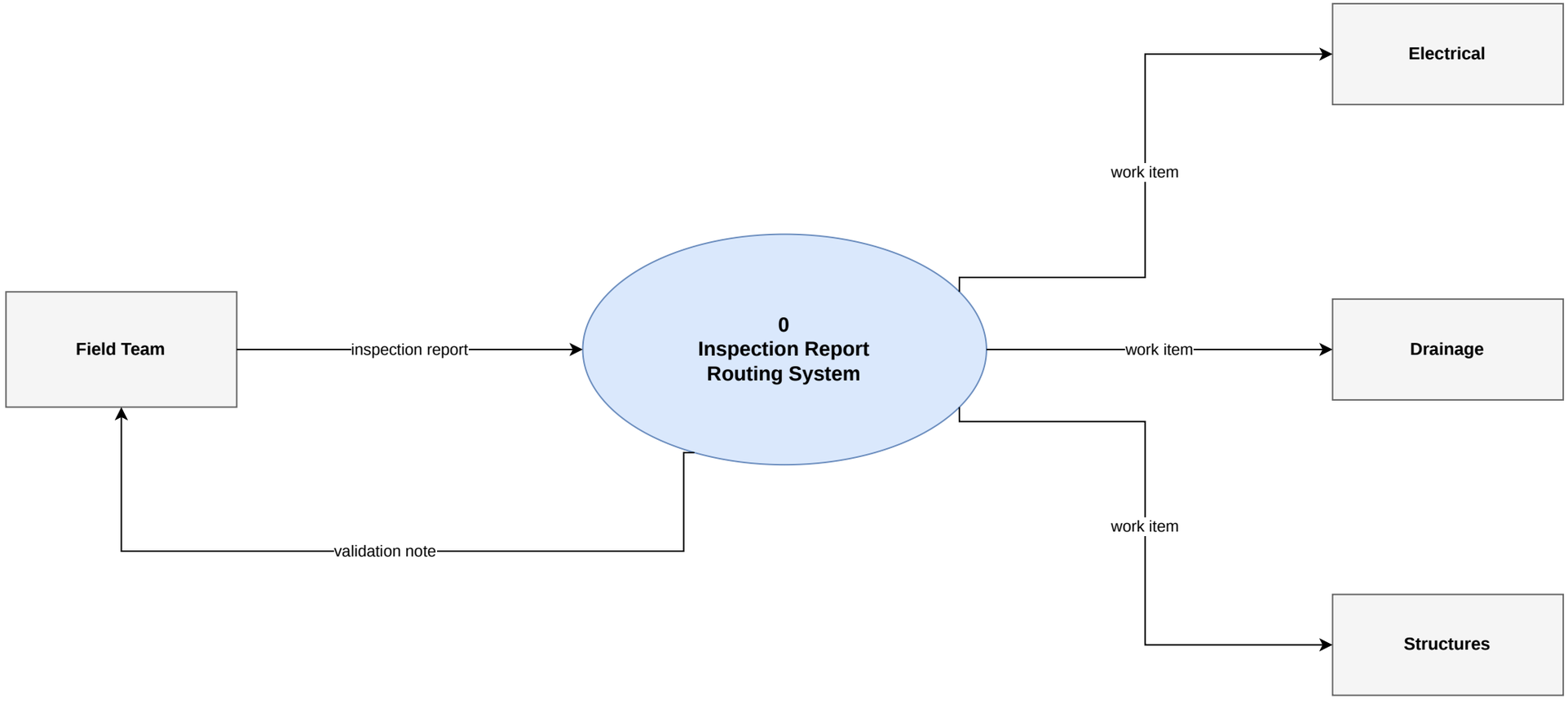
Page 1 — System Boundary (Context Diagram)

Inspection Report Routing System · Cedar Junction civic operations · source W06-S01

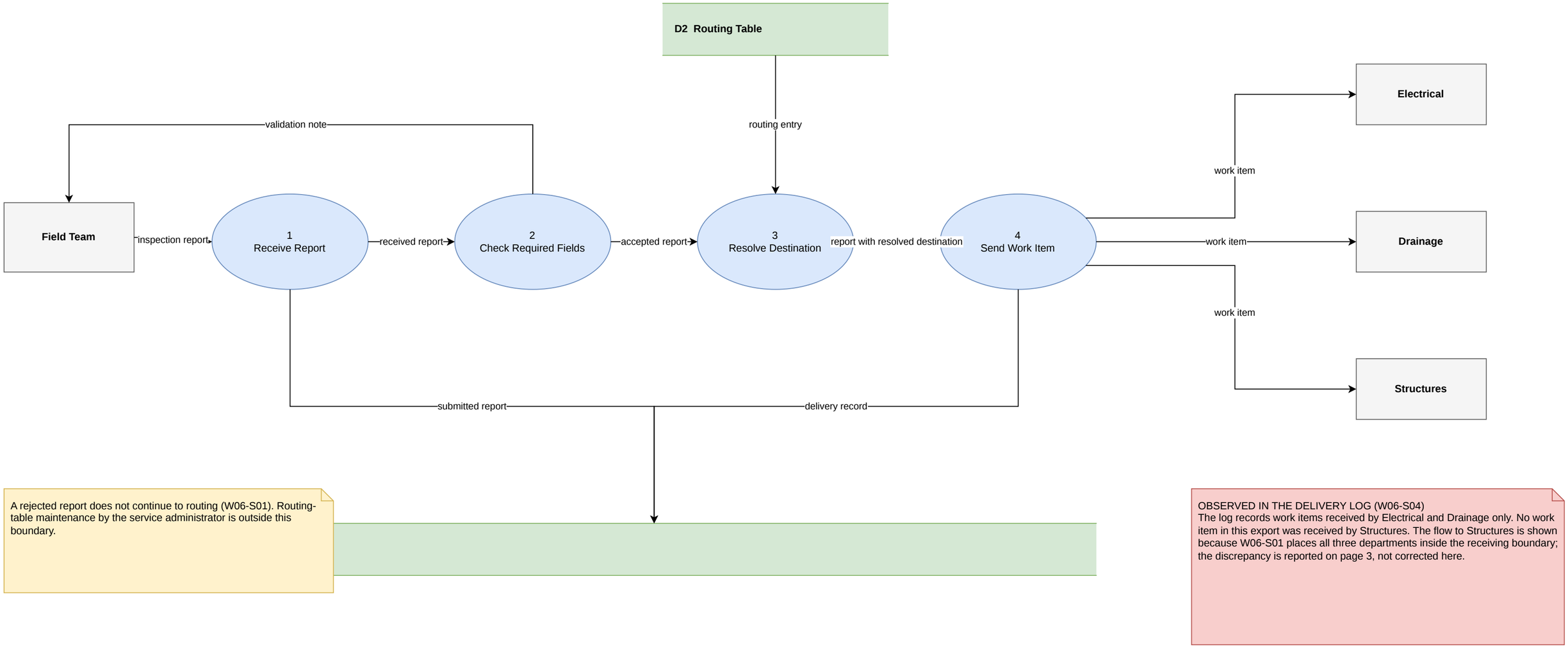


OUTSIDE THE BOUNDARY (W06-S01)
Flight control, physical inspections, repair work, and routing-table maintenance by the service administrator.

BALANCING
The external flows on this page are the same four flow types shown on page 2: inspection report in, validation note out, and work items out to the three departments.

Page 2 — Data Movement (First Breakdown of Process 0)

Processes, stores and labeled data flows inside the Inspection Report Routing System · source W06-S01



A rejected report does not continue to routing (W06-S01). Routing-table maintenance by the service administrator is outside this boundary.

OBSERVED IN THE DELIVERY LOG (W06-S04)
The log records work items received by Electrical and Drainage only. No work item in this export was received by Structures. The flow to Structures is shown because W06-S01 places all three departments inside the receiving boundary; the discrepancy is reported on page 3, not corrected here.

Page 3 — Data Definitions and Finding

Curtis Loop · CIT 140 Week 6 · Inspection Report Routing System

```
1. Data definitions — inspection report (field contract, W06-S02)

report_id Unique report reference; CJ- followed by three digits.
asset_id Reference for the inspected asset; two-letter prefix and three digits.
asset_type One of STREETLIGHT, DRAIN or BRIDGE.
finding_code One of LAMP_OUT, BLOCKED or SURFACE_DAMAGE.
priority One of ROUTINE or URGENT; urgency does not change the department.
captured_at UTC capture time, YYYY-MM-DDTHH:MM:SSZ.
photo_ref Attachment reference, IMG- followed by three digits.
survey_area One of NORTH, CENTRAL or SOUTH.

All eight fields are required. Process 2 checks the report against this contract.
```

```
2. Data stores

D1 Report Register Submitted reports written by process 1 and delivery records written by process 4.
The delivery log in W06-S04 is an export from this store.
D2 Routing Table Routing entries read by process 3 to resolve a destination.
Maintained by a service administrator outside the modeled boundary.
```

```
3. Required destination policy (W06-S03)

STREETLIGHT → Electrical DRAIN → Drainage BRIDGE → Structures

ROUTINE and URGENT set urgency within the receiving department. They do not change the department.
```

```
4. Required against observed (W06-S02, W06-S03, W06-S04)

report_id asset_type required received by result
CJ-601 STREETLIGHT Electrical Electrical match
CJ-602 DRAIN Drainage Drainage match
CJ-603 BRIDGE Structures Electrical DISCREPANCY
CJ-604 BRIDGE Structures Electrical DISCREPANCY
CJ-605 STREETLIGHT Electrical Electrical match
CJ-606 DRAIN Drainage Drainage match
CJ-607 BRIDGE Structures Electrical DISCREPANCY
CJ-608 STREETLIGHT Electrical Electrical match

5 of 8 delivered to the required department. All three BRIDGE reports went to Electrical.
Structures received no work item in this export.
```

5. Finding

Every report in the export was validated as ACCEPTED, so the field contract check in process 2 is not the point of failure. The discrepancy appears after validation, at the point where process 3 resolves a destination: all three BRIDGE reports were delivered to Electrical instead of Structures, while STREETLIGHT and DRAIN reports reached their required departments.

The pattern is consistent rather than intermittent, and it tracks asset_type alone. It is not explained by priority, since CJ-604 was URGENT and CJ-603 and CJ-607 were ROUTINE and all three were routed the same way, and W06-S03 states urgency does not change the department. It is not explained by survey_area, since the three affected reports came from CENTRAL, NORTH and SOUTH.

Effect on the receiving departments: Structures received no inspection work from this export, including a BRIDGE report marked URGENT, while Electrical received three work items outside its remit.

6. What the records do not establish

The cause of the discrepancy is not established by the supplied evidence. W06-S03 describes the required destination but does not prove what entries the running Routing Table contained, and no runtime table snapshot, deployment history or code was supplied. W06-S04 states that a routing discrepancy is evidence of a result and that its technical cause remains a question until supporting evidence is obtained.

Open questions for the next step: what entry, if any, D2 Routing Table holds for BRIDGE; when that entry was last changed and by whom, which falls to the service administrator outside this boundary; and whether any earlier export shows BRIDGE reports reaching Structures. This finding reports the discrepancy and does not record a repair.